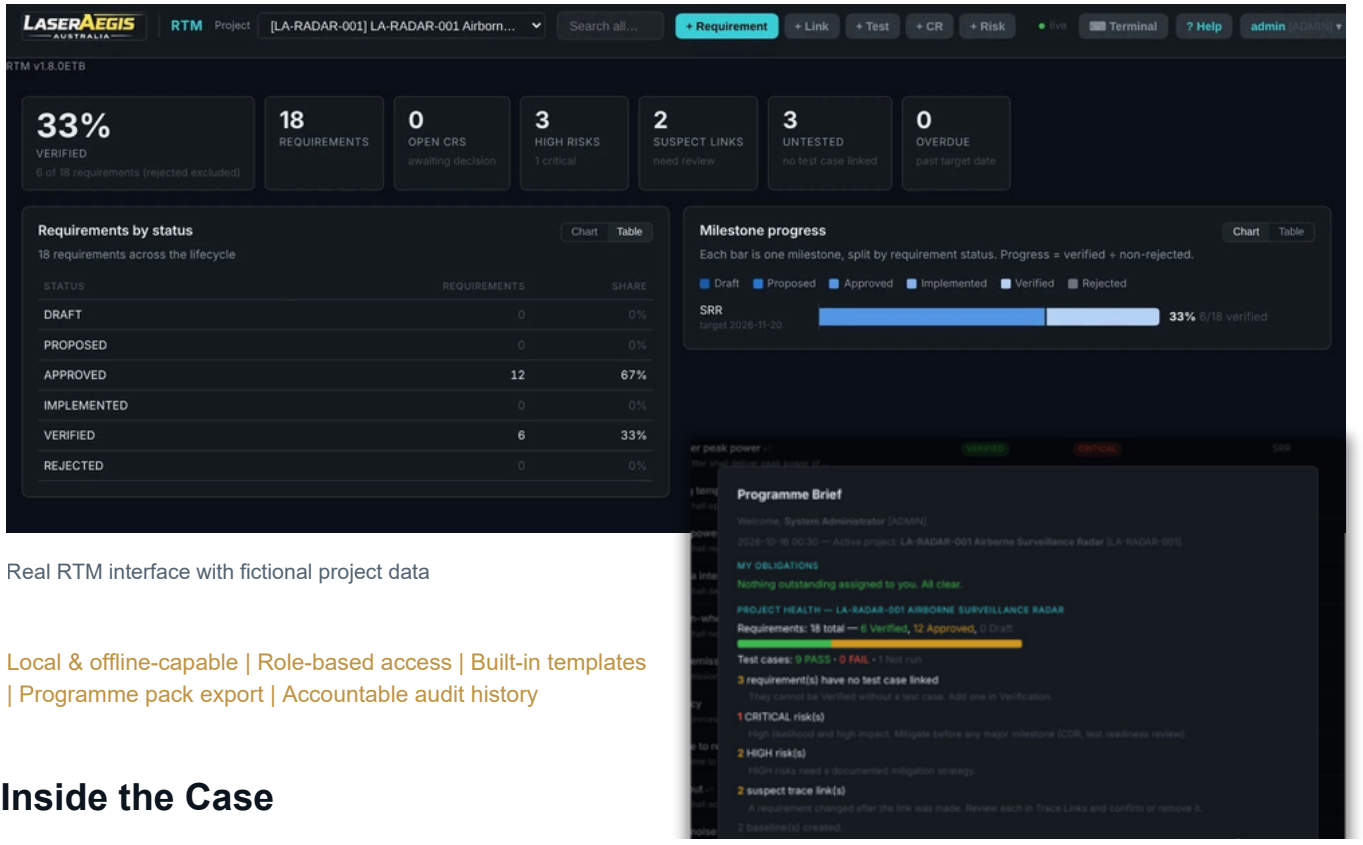
ENGINEERING CASE STUDY IN RTM

From Supplier Answer To Engineering Evidence

An Airborne Surveillance Radar Project in RTM

Follow this radar case to see **how RTM connects customer requirements, supplier commitments and controlled changes with verification results and supporting evidence** - so your team spends less time reconciling files and more time resolving engineering issues.



Real RTM interface with fictional project data

Local & offline-capable | Role-based access | Built-in templates | Programme pack export | Accountable audit history

Inside the Case

Less supplier follow-up: See the agreed answer, its conditions and earlier decisions together.

Controlled design changes: Follow a power shortfall through its system impact, approval and dependency review.

Clearer review preparation: Find evidence gaps with 24 report, check and export options and an eight-file Programme Pack.

Illustrative engineering scenario. Programme, suppliers, people and performance values are fictional. This case demonstrates RTM workflows, not measured customer outcomes or a completed acceptance claim.

BEFORE THE CASE

When Spreadsheets Stop Being Enough

Requirements Traceability Manager (RTM) is locally run requirements-management and traceability software for subsystem suppliers, systems engineers, integration and verification teams, and engineering consultancies.

Spreadsheets can work well while a project is simple. The burden grows when supplier replies change, approved requirements affect other work and review evidence sits in separate files. RTM keeps those relationships together, reducing repeated cross-file reconciliation as the project changes.

Capture
Source clauses and requirements

Control
Links, reviews and changes

Exchange
Supplier replies and decisions

Verify
Tests, documents and outputs

Why a team would choose RTM

Capability	Practical value for the team
Keep control of project data	Local, offline-capable operation; no separate central or licence server to configure
Start with less setup	Use templates and CSV / ReqIF exchange; check mapping and links.
Reduce supplier friction	Retain response rounds, clarifications and exception decisions. Suppliers can reply by CSV without running RTM.
Protect accountability	Role permissions control actions; versions and recorded audit history retain the trail
Reduce review assembly	24 report, check and export options, including an eight-file Programme Pack

HOW TO READ THIS CASE

Follow supplier negotiation, design change, a failed test and evidence review. Each stage shows where RTM reduces coordination work or makes an engineering risk easier to see.RTM was developed through LaserAegis' internal defence RF and laser engineering work, with experience across seven internal projects. The fictional case shows the workflow, rather than measured customer results.

01 / THE SCENARIO

Small Changes Project-Wide Consequences

An airborne surveillance radar team must turn customer invitation to tender (ITT) clauses into supplier commitments and verified requirements. Its challenge is keeping the agreed position clear when a supplier falls short or the design changes.

Project element	In this example
Source obligations	12 tender clauses, ITT-001 to ITT-012
Engineering requirements	18 records, LA-001 to LA-018, covering performance, interfaces, safety, power and maintainability
Subsystem suppliers	Acme Microwave: transmitter. Orbital Software: signal processing. Kestrel Power Systems: power subsystem
Review milestones	SRR: System Requirements Review. PDR: Preliminary Design Review. CDR: Critical Design Review. A baseline preserves the requirement state at a milestone

Three questions drive the case

Can the processor meet the timing limit? Orbital initially offers 120 ms against a 100 ms requirement. The team needs the reply history, exception decisions and final conditions.

What if the transmitter cannot meet its rating? Acme offers 8 kW instead of 10 kW. A system-level range requirement is affected; changing only the supplier's row would lose the wider decision.

What can the team support at review? A safety-related test fails before a correction and retest. Other tests and documents are unfinished, so a single "complete" label would be misleading.

> THE VALUE TO FOLLOW
>
> A list of requirements alone cannot answer these questions. RTM connects the obligation, supplier decision, approved change and evidence, helping the team resolve issues without repeatedly reconstructing the project history.

02 / CAPTURE AND CONNECT

Give Every Decision A Requirement To Return To

Before coordinating suppliers, the team gives each requirement a stable ID, owner, source and verification approach in RTM. Built-in templates provide the starting structure. These four records become the anchors for the decisions that follow.

Record	Initial requirement focus	Why it matters
LA-001	Detect a 1 m^2 target at 100 km, with Pd >= 0.9 and Pfa <= 1e-6.	System performance affected by the transmitter change.
LA-003	At least 10 kW peak transmitter power across 9.0 to 9.5 GHz.	Acme's proposed hardware falls short.
LA-007	No radiation while the weight-on-wheels signal is asserted.	Ground-safety behaviour must be tested.
LA-009	Deliver plots to the tracker within 100 ms of dwell completion.	Orbital's reply requires several rounds.

Structure and dependency answer different questions

LA-001 traces to source clause ITT-001. Receiver noise figure (LA-012) and antenna gain (LA-013) are derived below LA-001. Transmitter power (LA-003) is a separate requirement that contributes to the range outcome. Typed trace links make that engineering relationship explicit.

Before the baseline, reviewers identify unclear wording: the range requirement needs its scan basis and the power requirement needs its duty cycle. The team revises the statements, records new review outcomes and then approves the requirements.

> **LESS RECONCILIATION. CLEARER TRACEABILITY**
>
> A reviewer can follow a requirement back to its source and forward to related work. Engineers spend less time matching IDs across files and have a clearer starting point for impact analysis. Templates reduce setup work while retaining project-specific fields.

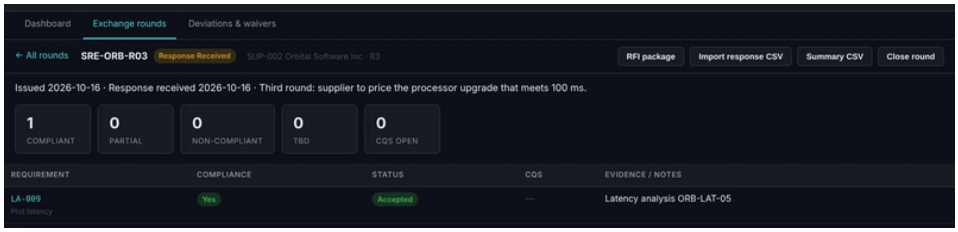
Initial wording is abbreviated for readability. Pd = probability of detection; Pfa = probability of false alarm. The optional reference workbook contains the full initial statements.

03 / SUPPLIER RESPONSES

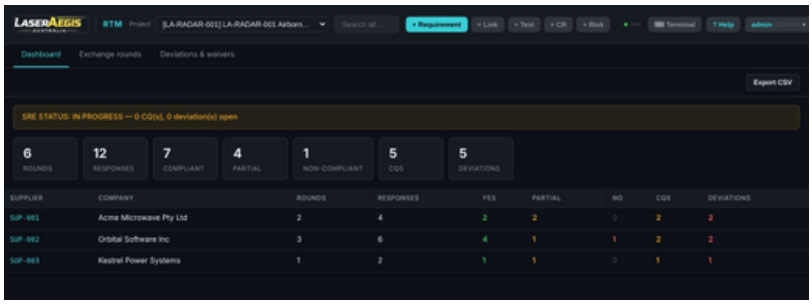
Stop Chasing The Supplier's Latest Answer

Orbital must meet LA-009: plot latency no greater than 100 ms. RTM keeps each issued package and returned CSV in a dated round, reducing the need to search email for earlier replies. Orbital uses exchange files without installing RTM.

Round	Supplier position	Decision recorded
01	120 ms	Reject deviation DEV-003: above the latency budget.
02	110 ms after optimisation	Reject deviation DEV-004: still above 100 ms.
03	100 ms with an added GPU card	Accept the position with the dependency, eight-week lead time and AUD 22,000 cost visible.



Third-round response: 100 ms accepted. Earlier rounds remain available; verification is recorded separately.



RTM supplier-response overview. Open-item counts and recorded totals describe different things.

Clarify the answer, then decide the exception

Questions and supplier answers stay with the response. Where the offer departs from the requirement, the team records a deviation or waiver and the rationale, conditions and decision. Acme's power shortfall is carried into the controlled-change stage rather than silently treated as compliant.

LESS CHASING. FEWER LOST CONDITIONS.

The final "yes" stays attached to its GPU dependency, cost and lead time. Teams can reduce repeated clarification and the risk of acting on an outdated reply, while retaining the rationale behind the agreed supplier position.

Supplier acceptance records an agreed position. It is separate from performing verification and reviewing the resulting evidence.

04 / CONTROLLED CHANGE

Change the Design Keep Control of the Impact

After the SRR baseline, Acme cannot deliver the initial 10 kW transmitter. Updating the power row alone would hide a system effect. The team assesses the range consequence and records two linked Engineering Change Orders (ECOs) as RTM change requests.

Record	SRR starting value	Approved case change
LA-003 / Peak power	10 kW	8 kW
LA-001 / Detection range	100 km	95 km



Selected baseline comparison: two changed requirements. The before-and-after statements remain connected to their recorded requirement state.

The decision continues beyond the edit

The team records the impact analysis, approval and applied change. Related trace links become suspect so engineers revisit their dependencies. A later baseline comparison identifies what changed; Acme's power deviation is resolved against the approved 8 kW requirement and a revised supplier response.

> **CONTROLLED CHANGE. ACCOUNTABLE DECISIONS.**
>
> Baseline comparison and suspect links direct attention to what changed and what needs review. Linked approvals and recorded audit history preserve accountability, helping engineers make changes without losing the decision or overlooking related work.

The exercise uses R = 100 x (8/10)^0.25, about 94.6 km, rounded to 95 km, assuming other radar-equation factors are unchanged. This is a simplified impact illustration, not a validated radar design. Admins have additional edit authority; those actions remain versioned and audited.

05 / VERIFICATION

Find Verification Gaps Before The Review

LA-007 prohibits radiation when weight-on-wheels indicates that the aircraft is on the ground. The first bench test fails: RF persists for 120 ms. RTM connects the result to the requirement, so the correction and retest can be followed in the same engineering context.

Stage	Case record	Meaning for the review
First test	FAIL: RF output persists for 120 ms.	The recorded behaviour does not satisfy the criterion.
Correction	Interlock debounce changed to 20 ms.	A change is made; the edit alone does not establish a pass.
Retest	PASS, with updated actual result, date and evidence reference.	The passing retest is recorded without erasing the earlier failure history.

The same project also contains unfinished work

Requirement	Recorded activity status	Next question
LA-001 / Range	PASS by analysis	Are the assumptions and supporting analysis current?
LA-004 / Temperature	BLOCKED	When will the thermal chamber be available?
LA-005 / Power-loss report	NOT_RUN	Has the planned demonstration been performed?

The project also links risks to requirements, including the hazard associated with a failed interlock. Likelihood, impact, mitigation, residual risk and responsibility help the team keep follow-up work visible.

FOCUSED FOLLOW-UP. CLEARER VERIFICATION.

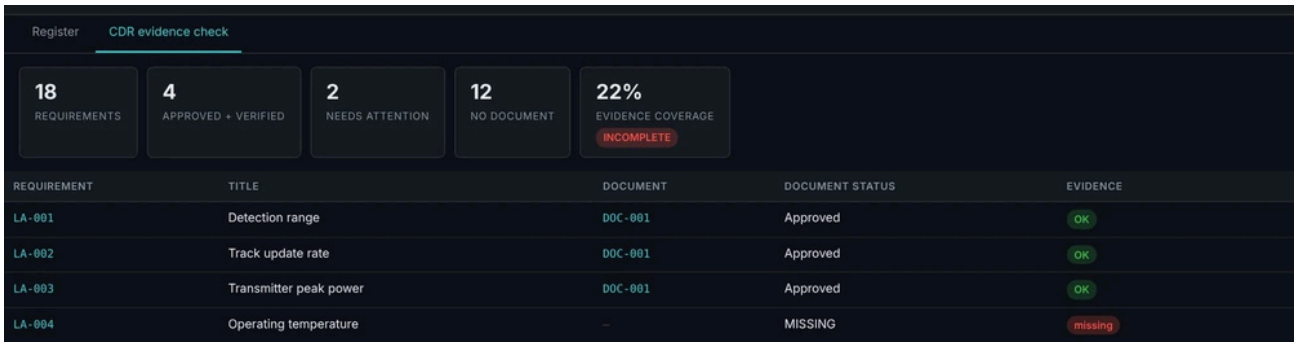
Linked test status helps the team focus follow-up on failed, blocked and unrun activities. Reviewers can see the criterion, result and evidence reference together, reducing manual cross-checking and the risk of mistaking progress for verification.

The 20 ms debounce is a case implementation value, not a universal safety limit or a substitute for project-specific acceptance criteria.

06 / EVIDENCE

Prepare Evidence Spend Less Time Chasing It

A passing test is difficult to defend with a missing or outdated report. RTM links document revision and approval to the requirement, helping the team find gaps before assembling a review pack.



Snapshot: 4 of 18 requirements have approved and verified document support (22%); 2 need attention and 12 have no document. This is document coverage, not a test pass rate.

Two changes that matter after registration

A report is superseded. Acme issues a new acceptance-test report revision. RTM retains the relationship between the replacement and the earlier document, so the team can identify the current supporting record.

A file changes. Orbital reissues its latency report with altered content. RTM detects the registered SHA-256 mismatch. The team reviews the change and can flag related requirements and trace links for another check.

DOC ID	TITLE	TYPE	REV	STATUS	FILE	LINKED REQUIREMENTS
DOC-001	System Specification	Specification	A	Approved	OK	LA-001 LA-002 LA-003
DOC-002	~~Transmitter ATP Report~~	Test Report	A	Superseded	OK	LA-003
DOC-003	Latency Report 05	Supplier Document	A	Approved	Changed	LA-009

The document register flags a changed file. The team decides how to review and reconcile the updated evidence.

> **LESS ASSEMBLY. STRONGER REVIEW PREPARATION.**
>
> Evidence checks and file-change detection help the team find weak support earlier. Built-in reports and the eight-file Programme Pack reduce repeated assembly of project outputs; engineers can focus on closing gaps and explaining the results.

Snapshots show different stages of the example, not one simultaneous final state. File hashes detect changed bytes; they do not certify the engineering conclusion or guarantee customer acceptance.

07 / PUT IT TO WORK

See Where Your Team Is Doing Avoidable Work

Choose one requirement in your own project. How many files, emails and people would you need to answer these questions? Use this checklist to identify where connected RTM records could reduce repeated coordination and review preparation.

Ask	Look for
1. What is the current agreed obligation?	Requirement ID, full statement, version, source clause and applicable baseline.
2. What did the supplier commit to?	Issued requirements, response round, answer, conditions and clarification history
3. Were exceptions decided?	Clarification outcome, deviation or waiver, rationale and responsible decision-maker.
4. Was the change assessed and authorised?	Impact assessment, approval and the applied requirement change.
5. Were dependencies revisited?	Suspect links and the review of affected requirements or activities.
6. What was verified, by which activity?	Method, criterion, actual result, date and retest history.
7. Is the supporting document current?	Linked document, revision, approval, supersession and integrity check.
8. What remains open for the review?	Blocked or unrun tests, missing documents and unresolved supplier positions.

WHAT THE EXAMPLE DEMONSTRATES

RTM is worth assessing when keeping these answers current has become a recurring team task. Its value is less reconciliation, clearer responsibility and repeatable review outputs, with local operation and CSV / ReqIF exchange for project handover.Which question is hardest to answer in your project? Tell us, and we can focus a guided demonstration on that workflow and the records your team needs.

Optional supporting data

The Excel reference and optional CSV contain all 18 initial requirements at the starting values (10 kW and 100 km). Use them to explore the example; they are reference data, not a runnable RTM project or a ready-to-import dataset.

MAKE THE NEXT STEP PRACTICAL

See How RTM Could Fit Your Project

Book a guided RTM demonstration

We will walk you through a complete project, from tender clauses and requirement structure to supplier responses, controlled changes, verification and review outputs. See how the main features work together, then tell us your main challenge and we'll focus on the workflow that matters most to your team. No project files are needed for an initial demonstration.

In the demonstration	What your team can assess
Capture and connect	How templates and CSV / ReqIF exchange provide a starting point.
Respond and control	How reply rounds, exceptions and approvals preserve the agreed position.
Verify and prepare	How tests, evidence checks and project outputs support the next review.
Deploy and govern	What local setup, data preparation, roles and ongoing upkeep involve.

Try a scope that fits your project

> Pilot plans can be tailored to your project scope, team size and duration, including short-term use. Start with a defined task and an agreed budget, then decide whether to expand. You can assess the value without committing to a wider rollout at the start.

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Prefer email or phone? Contact xander@laseraegis.dev or +61 451 986 931. Tell us your main project challenge and time zone.